

Practical Data Analysis Tips Pt. 1

ECON 490

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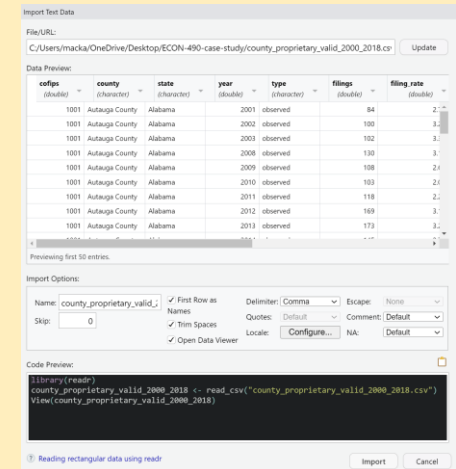
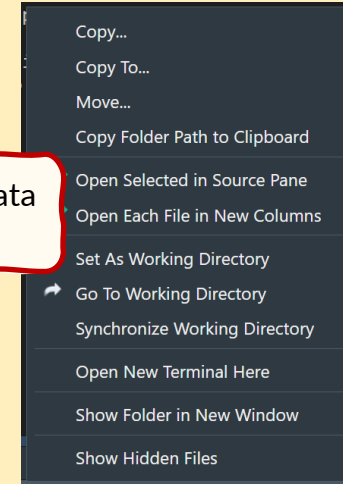
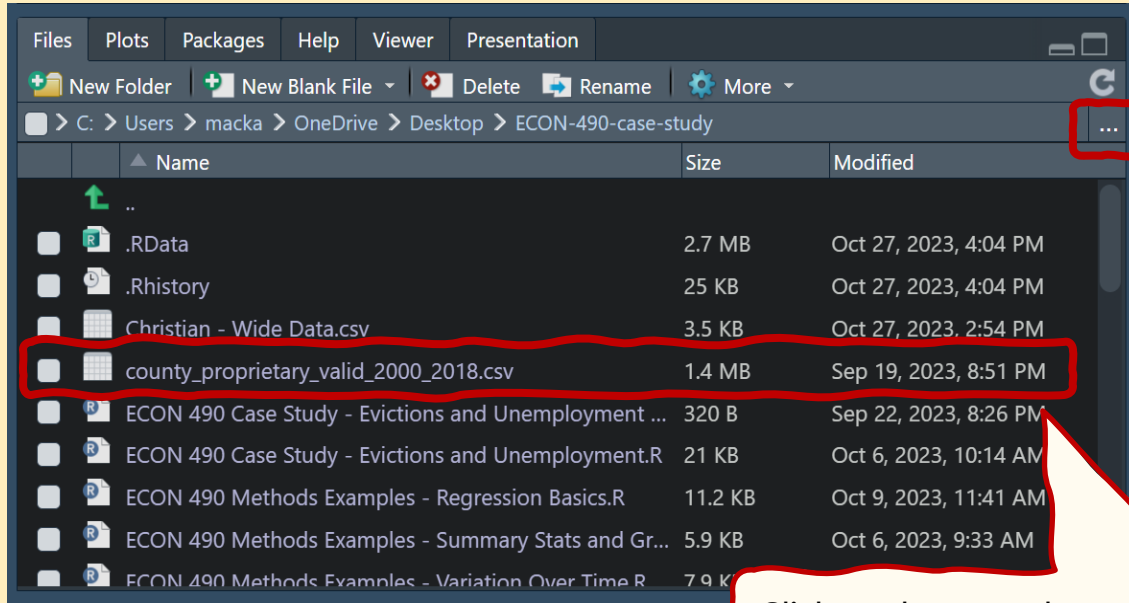
Slides Overview

In these slides, we'll discuss useful data analysis tools/concepts:

- Data structure terminology
- Merging and appending data sets
- Using squared terms and logs in regressions

Loading Data into R

Click ..., load folder with your data and set working directory



Click on data set, choose "Import Data," to preview & load data

Wide vs. Long-Formatted Data

Two types of data structures:

- **Wide** data has same variable in multiple columns (here, GDP is split by year)
- **Long** data has each variable in one column

In almost all cases, data should be in long format for running regressions

- In R, use `pivot_longer()`

	state	year	GDP
1	AZ	2020	100
2	AZ	2021	105
3	AZ	2022	107
4	CA	2020	157
5	CA	2021	163
6	CA	2022	168
7	NM	2020	95
8	NM	2021	102
9	NM	2022	103

Long-formatted data

	state	gdp.2020	gdp.2021	gdp.2022
1	AZ	100	105	107
2	CA	157	163	168
3	NM	95	102	103

Wide-formatted data

Using AI Effectively

Many students don't use AI effectively because they don't provide enough context

- However much information you've given AI, give more!
- For data/coding questions, this means letting AI “see” your data

In R, the `str()` function should be your best friend

General Workflow: Start by explaining desired output *in detail*

- Explicitly tell AI what packages to use (e.g., tidyverse/estimatr)
- Show AI your data using `str()` and regression output using `summary()`

Combining Data Sets

Merging data == combining data sets – why do we need to do this?

- For your projects (and most data analysis), you'll get data from **multiple** sources
- You combine data sets by **merging or joining** them together

Examples include combining:

- State-level data with individual-level survey responses
- Location-based crime data with local area demographic data
- NBA player performance data with salary and contract records

The Shape of Data Sets

When you merge two data sets, your data expands **horizontally**

- You started with some set of columns/variables, then added new columns
- Different variables across data sets

Sometimes, you have data that you want to **stack or append**

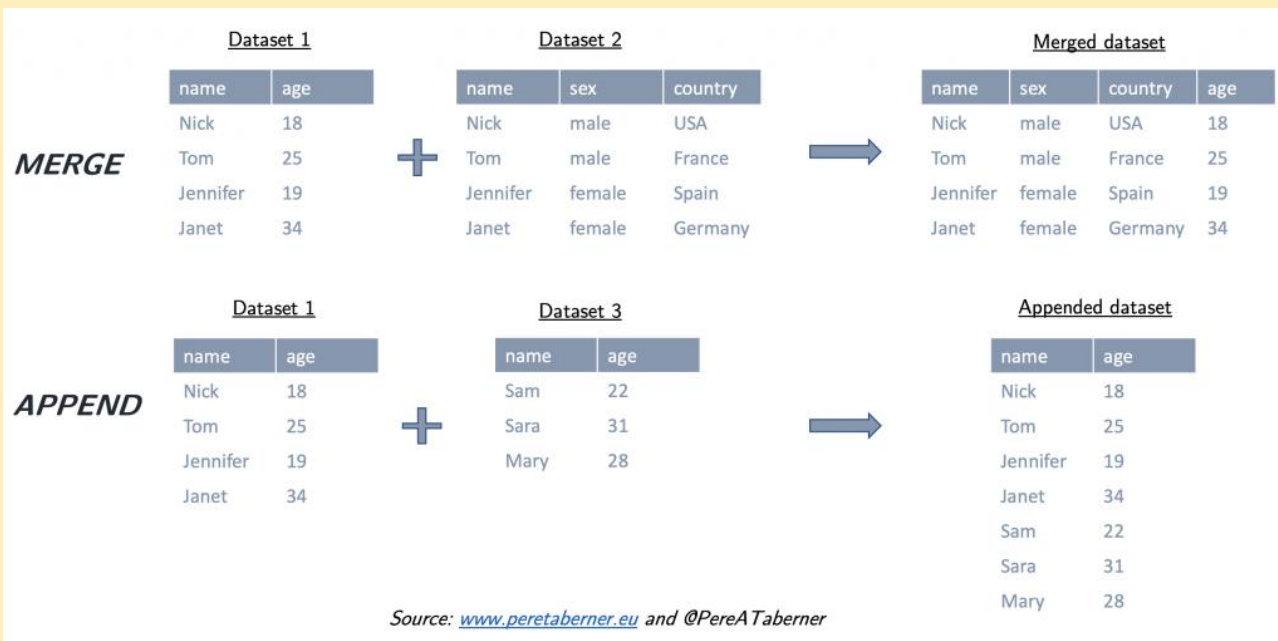
- Example: county-level pollution data for CA in 2020 and 2021
- **Same** variables in **both** data sets → append data to get panel with both years

Appending data expands your data **vertically**

- Depending on context, merges might also, but main goal = new columns

Appending Data in R

In screenshot below, use `bind_rows()` to **append** data sets 1 & 3



Before You Combine Data

Before you combine data sets, you need to understand ***data structure***

- Within both data sets, what variables uniquely identify each row?

Across both data sets, what variables will you use to link data?

- These are your “key” or “by” variables
- What is the shared structure of both data sets? Defined by ***less*** granular data

Example: Combining individual-level CPS with state-level home price data

- While CPS has individual identifiers, the ***shared structure*** is state and year, so these are the key or by variables (state-level data is less granular)

Preparing Data for Merging

Once you've identified your key variables, make sure they're **consistent**

- Across both data sets, is each variable the **same format** (e.g., factor, string, etc.)?
- Are spellings consistent? E.g., check “California” vs. “california” vs. “CA”

If necessary, create new, consistent versions of key variables using `mutate()`

Check you don't have duplicates across key variables

- E.g., in state-by-year ACS data, suppose you found 2 Arizona-in-2010 rows
- Drop extra row using `filter()` or combine via `group_by()` + `summarize()`

Join Functions in R

Given two data sets, you'll have one of the following types of merges or joins:

- **1-to-1**: Each observation in 1st dataset matches exactly one row in 2nd
- **Many-to-1**: Multiple observations in 1st dataset match to same row in 2nd
- **Many-to-Many**: Multiple rows match to multiple rows → avoid doing this!

Most useful join functions in R for your projects:

- `inner_join()`: Keep **only** matched rows in **both** data sets
- `left_join()`: Keep all rows from 1st data set and all matches from 2nd
- `full_join()`: Keep all rows from **both** data sets

After Merging Data

The most important thing to do is “sanity check” your new merged data set

- Check the number of rows in the matched data set using `nrow()`
- Does it match what you expected?

Check NAs and summary statistics for important variables using `summary()`

Example: Combine state-level crime data with state-level ACS via `inner_join()`

- This keeps matched rows, so you should have rows for 50 states plus DC multiplied by 5 years; if you don't, something went wrong!

Join Examples

Inputs

DF1

ID	Value
A	123
B	769
C	475
D	978

DF2

ID	Level
A	Red
B	Blue
E	Green
F	Yellow

Join Functions

```
inner_join(DF1, DF2,  
           by = "ID")
```

```
left_join(DF1, DF2,  
          by = "ID")
```

Outputs

ID	Value	Level
A	123	Red
B	769	Blue

ID	Value	Level
A	123	Red
B	769	Blue
C	475	NA
D	978	NA

Non-Linearity

We can use OLS to estimate regression equations like the following:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + u$$

This lets Y be a function of X **and** X squared

OLS requires equations that are “linear in parameters”

- In other words, equal to the sum of intercept + variables multiplied by β 's
- OLS is ***flexible*** w.r.t. the functional form of individual explanatory variables

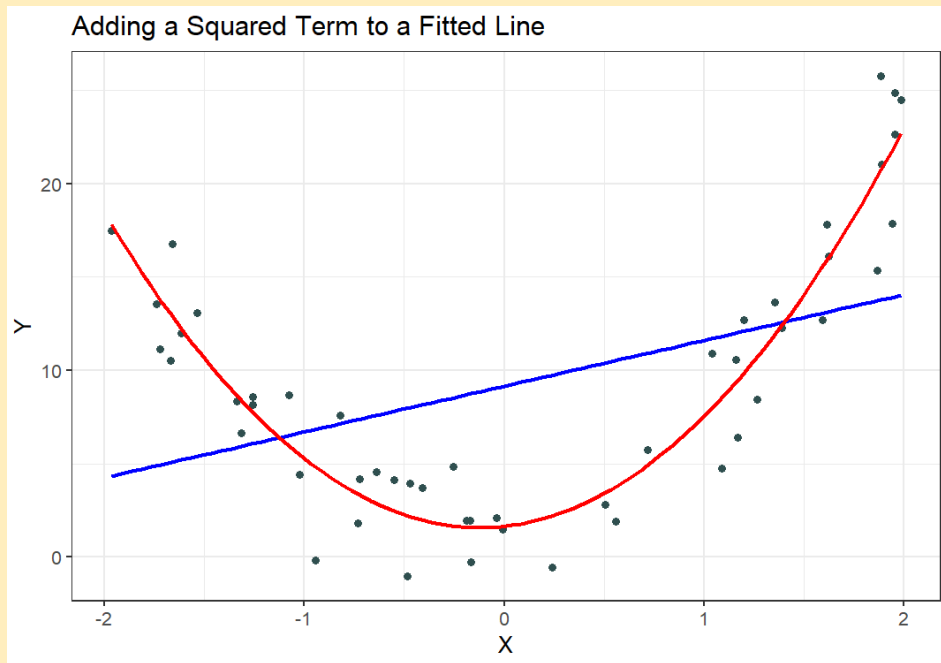
Squared Term Example

Equation for the blue line:

$$Y = \beta_0 + \beta_1 X + u$$

Equation for the red line:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + u$$



(Natural) Logs

Another common data transformation is using the natural log function

- Widely used in metrics + stats, generally just referred to as log
- In R, calculated using the `log()` function

When using logs, we'll refer to the **level** of a variable X , and the log value $\log(X)$

- To calculate the log of X , note that X must be strictly greater than 0

Using logs in regression can make coefficients easier to interpret

Logs in Regressions

Suppose we want to estimate the relationship between home sizes and sales prices:

$$\text{Home Price}_i = \beta_0 + \beta_1 \text{Square Footage (SqFt)}_i + u_i$$

If we use **levels** of both variables, we use “raw” home price and sq. footage variables

- Suppose $\hat{\beta}_1 = 120 \rightarrow 1\text{-sq. foot } \uparrow \text{ in size associated with a \$120 increase in price}$

If we use log values of both variables, the (new) resulting $\hat{\beta}_1$ is an **elasticity**

- Write new equation as: $\ln(\text{Home Price}_i) = \beta_0 + \beta_1 \ln(\text{Square Footage (SqFt)}_i) + u_i$
- Suppose $\hat{\beta}_1 = 0.85 \rightarrow 1\% \text{ increase in size is associated with a 0.85\% increase in price}$

Percentage vs. Percentage Points

Related to logs, an important distinction you should know

- **Percentage points** = difference between two percentages

Suppose an NBA player makes 40% of their 3's in year 1 and 50% in year 2

- Did their 3-point shooting percentage increase by 10%? **No!**

How to correctly characterize this change?

- Their shooting percentage increased by 25% $\rightarrow 100\% \times (50\% - 40\%) / 40\% = 25\%$
- Their shooting percentage increased by 10 **percentage points** (p.p.) $\rightarrow 50\% - 40\% = 10$ p.p.

Practice Problems

For which of the following variables can we use the $\log()$ function?

- Age in years for a sample of adults which is always greater than or equal to 18
- Education which is a 3-level factor variable (defined using ACS sample data definitions)
- Household income which includes values less than \$0 due to small business losses

Suppose the inflation rate was 4% in year 1 and 6% in year 2. We can say that:

1. The inflation rate increased by 50% $\rightarrow 100\% \times (6\% - 4\%) / 4\% = 50\%$
2. The inflation rate increased by 2 **percentage points** $\rightarrow 6\% - 4\% = 2 \text{ p.p.}$